Modbus

Industrial Control Systems · SCADA Hub

COURSE TOPICS

1. **Introduction:** Protocol origins, the 1979 register model, and why the original design persists unchanged in modern field devices.
2. **Data Model:** Coils, discrete inputs, input registers, and holding registers — table structure, data types, and access rules.
3. **RTU Mode:** RS-485 framing, CRC-16 integrity, multi-drop bus wiring, and inter-frame silence on serial Modbus links.
4. **ASCII Mode:** Hex-encoded framing, colon delimiter, overhead comparison to RTU, and legacy HMI equipment identification.
5. **Modbus TCP:** Port 502, the MBAP header, transaction ID pipelining, and security implications of Ethernet transport without authentication.
6. **Function Codes:** Read, write, and diagnostic codes; the subset covering ninety-five percent of deployments; error response handling.
7. **Exception Codes:** Eight exception identifiers, meanings, and systematic diagnosis from exception response to root cause.
8. **Data Types:** Register-pair conventions, byte order, word order, and floating-point encoding across vendor implementations.
9. **Troubleshooting:** Addressing offset errors, byte-order mismatches, RS-485 termination faults, firewall drops, and poll rate overruns.
10. **Security:** Protocol-level limitations and compensating controls: network segmentation, unidirectional gateways, and anomaly detection.
11. **Practice Lab:** Register mapping, Exception Code 02 tracing, serial-to-TCP gateway configuration, byte-order diagnosis, and Wireshark traffic analysis.

LEARNING OUTCOMES

- Configure Modbus RTU and TCP device connections from device datasheet parameters.
- Read and interpret function code and exception code responses in a traffic capture.
- Apply byte-order and register-pair conventions to map floating-point and 32-bit values.
- Diagnose serial and TCP-layer Modbus faults using systematic isolation procedures.
- Specify network-layer compensating controls appropriate for a Modbus deployment.

Certification Relevance

- **ISA CCST** — Modbus appears in industrial communications and data acquisition domains at all three CCST levels.
- **GICSP (GIAC)** — OT protocol knowledge and Modbus security posture are tested throughout ICS security exam domains.
- **CompTIA CySA+** — Industrial network traffic analysis including Modbus TCP anomaly detection.

Next Steps

1. Take Modbus before OPC UA. The data model concepts carry over directly.
2. Download ModRSsim2 (free) and Modbus Poll (trial) to run register reads against a live simulator.
3. Review IEC 61158 for the formal specification behind the protocol details.
4. ISA CCST Level I is the right entry credential for industrial automation work.